DERIVATIONS

Compressible Flow & Gas Dynamics

Nozzle Flow · Prandtl–Meyer Expansion · Shock–Expansion Theory · Rayleigh Flow

Compiled from the course PDFs in this repository:
Lecture 22 · Nozzle Flows and Problems · Unit 3_A to 3_E · Unit 4_Rayleigh Flow

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Common Notation

Symbols used throughout. Each derivation also lists the meaning of every symbol under its formulas (“where, …”).

γ	=	ratio of specific heats, c_p/c_v (1.4 for air)
R	=	specific gas constant (287 J/kg·K for air)
c_p	=	specific heat at constant pressure (1005 J/kg·K for air)
$p,\ T,\ \rho$	=	static pressure, static temperature, density
$p_0,\ T_0,\ \rho_0$	=	stagnation (total) pressure, temperature, density
$p^*,\ T^*,\ \rho^*$	=	values at the sonic ($M = 1$) condition
$u,\ V$	=	flow velocity
a	=	speed of sound, $\sqrt{\gamma RT}$
M	=	Mach number, V/a
$A,\ A^*$	=	local area; sonic throat area
μ	=	Mach angle, $\sin^{-1}(1/M)$
$\theta,\ \beta$	=	flow deflection angle; oblique shock wave angle
$\nu(M)$	=	Prandtl–Meyer function
$\alpha,\ \varepsilon$	=	angle of attack; wedge half-angle of the diamond airfoil
$c,\ t,\ l$	=	chord, thickness, side length of an airfoil
$L,\ D$	=	lift and drag per unit span
$C_l,\ C_d$	=	lift and drag coefficients
q	=	heat added per unit mass (Rayleigh flow) / q_∞ = dynamic pressure

PART A

Quasi-One-Dimensional Nozzle Flow

Source: Lecture 22.pdf

A1. Governing Equations of Quasi-1D Flow (Differential Form)

Source: Lecture 22.pdf

Given: One-dimensional flow has constant area (A = const). Quasi-1D flow has $A = A(x)$ and all properties vary only with x. The integral (finite) forms between sections 1 and 2 are:

$$\rho_1 u_1 A_1 = \rho_2 u_2 A_2$$

where,

ρ = density of the gas

u = flow velocity

A = local cross-sectional area of the duct

$1, 2$ = inlet and exit sections

$$p_1 A_1 + \rho_1 u_1^2 A_1 + \int_{A_1}^{A_2} p \, dA = p_2 A_2 + \rho_2 u_2^2 A_2$$

where,

p = static pressure

$\int p \, dA$ = axial force exerted by the sloping duct walls on the gas

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

where,

h = specific enthalpy (= $c_p T$ for a calorically perfect gas)

$u^2/2$ = kinetic energy per unit mass

Step 1: Continuity – take a thin slice of thickness dx (properties change from ρ, u, A to $\rho + d\rho,\ u + du,\ A + dA$)

$$\rho u A = (\rho + d\rho)(u + du)(A + dA)$$

where,

$d\rho,\ du,\ dA$ = infinitesimal changes across the slice

Neglecting products of two differentials (second-order terms):

$$\boxed{d(\rho u A) = 0}$$

where,

$d(\rho u A)$ = change in mass flow rate, which must be zero for steady flow

Step 2: Momentum – net force on the slice = rate of change of momentum. The wall exerts $p\,dA$ on the gas

$$pA + p\,dA - (p + dp)(A + dA) = \rho u A \left[(u + du) - u\right]$$

where,

pA = pressure force on the inlet face

$(p + dp)(A + dA)$ = pressure force on the exit face

$p\,dA$ = pressure force on the sloping side wall

$\rho u A$ = mass flow rate through the slice

Neglecting $dp\,dA$ gives $-A\,dp = \rho u A\,du$, i.e.

$$\boxed{\frac{dp}{\rho} + u\,du = 0} \qquad \text{(compressible Bernoulli equation)}$$

where,

dp/ρ = pressure-work term

$u\,du$ = change of kinetic energy per unit mass

Step 3: Energy – for an adiabatic flow with no work

$$\boxed{dh + u\,du = 0}$$

where,

dh = change in specific enthalpy

A2. Area–Velocity Relation

Source: Lecture 22.pdf

To find: how the duct area A must change to change the flow velocity u in subsonic and supersonic flow.

Step 1: Continuity equation in logarithmic-differential form

$$d(\rho u A) = 0 \;\Rightarrow\; \frac{d\rho}{\rho} + \frac{du}{u} + \frac{dA}{A} = 0 \qquad \cdots (1)$$

where,

$d\rho/\rho$ = fractional change in density

du/u = fractional change in velocity

dA/A = fractional change in area

Step 2: Momentum equation, written in terms of $d\rho/\rho$

$$\frac{dp}{\rho} = -u\,du \;\Rightarrow\; \frac{dp}{d\rho}\,\frac{d\rho}{\rho} = -u\,du$$

For isentropic flow $dp/d\rho = a^2$:

$$a^2\frac{d\rho}{\rho} = -u\,du \;\Rightarrow\; \frac{d\rho}{\rho} = -\frac{u^2}{a^2}\,\frac{du}{u}$$

where,

a = speed of sound, $a = \sqrt{dp/d\rho}$ (isentropic)

$$\frac{d\rho}{\rho} = -M^2\,\frac{du}{u} \qquad \cdots (2)$$

where,

M = Mach number, $M = u/a$

Step 3: Substitute (2) in (1)

$$-M^2\frac{du}{u} + \frac{du}{u} + \frac{dA}{A} = 0$$

$$\boxed{\frac{dA}{A} = (M^2 - 1)\,\frac{du}{u}}$$

where,

dA/A = fractional change in duct area

du/u = fractional change in flow velocity

$M^2 - 1$ = sign of this term decides the duct shape

Step 4: Interpretation

Flow	**Sign of** (M^2-1)	**Relation**	**Conclusion**
Subsonic $0 \le M < 1$	negative	$dA/A = (-)\,du/u$	$u\uparrow$ needs $A\downarrow$ (convergent duct); $u\downarrow$ needs $A\uparrow$ (divergent duct)
Supersonic $M > 1$	positive	$dA/A = (+)\,du/u$	$u\uparrow$ needs $A\uparrow$ (divergent duct); $u\downarrow$ needs $A\downarrow$ (convergent duct)
Sonic $M = 1$	zero	$dA/A = 0 \Rightarrow dA = 0$	Minimum area: the **throat**

Step 5: Incompressible limit ($M \to 0$)

$$\frac{dA}{A} = -\frac{du}{u} \;\Rightarrow\; \frac{dA}{A} + \frac{du}{u} = 0 \;\Rightarrow\; d(Au) = 0 \;\Rightarrow\; Au = \text{constant}$$

where,

Au = volume flow rate (constant for incompressible flow)

> **Note:** Supersonic nozzle: $M < 1 \to M = 1$ at throat $\to M > 1$ (converging–diverging). Supersonic diffuser: $M > 1 \to M = 1$ at throat $\to M < 1$.

A3. Area–Mach Number Relation (Isentropic Flow Through a C-D Nozzle)

Source: Lecture 22.pdf

To find: the local Mach number M at any section of area A, in terms of the sonic throat area A^* (the area where $M = 1$).

Step 1: Continuity between the section (A) and the sonic section (A^*)

$$\rho^* u^* A^* = \rho u A \;\Rightarrow\; \frac{A}{A^*} = \frac{\rho^* u^*}{\rho u} = \frac{\rho^*}{\rho_0}\,\frac{\rho_0}{\rho}\,\frac{a^*}{u}$$

where,

$*$ = value at the sonic ($M = 1$) section; $u^* = a^*$

ρ_0 = stagnation density (constant in isentropic flow)

Step 2: Square it and use the isentropic relations

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{1}{\gamma - 1}}, \qquad \frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma + 1}\right)^{\frac{1}{\gamma - 1}}$$

where,

γ = ratio of specific heats, c_p/c_v

Step 3: Velocity ratio from the definition of the characteristic Mach number M^*

$$\left(\frac{u}{a^*}\right)^2 = M^{*2} = \frac{\frac{\gamma+1}{2} M^2}{1 + \frac{\gamma-1}{2} M^2}$$

where,

M^* = u/a^*, characteristic Mach number

Step 4: Combine

$$\left(\frac{A}{A^*}\right)^2 = \left(\frac{2}{\gamma+1}\right)^{\frac{2}{\gamma-1}} \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{2}{\gamma-1}} \frac{1 + \frac{\gamma-1}{2} M^2}{\frac{\gamma+1}{2} M^2}$$

The powers of the bracketed terms combine as $\frac{2}{\gamma-1} + 1 = \frac{\gamma+1}{\gamma-1}$, so

$$\boxed{\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2} M^2\right)\right]^{\frac{\gamma+1}{\gamma-1}}}$$

where,

A/A^* = area ratio (local area / sonic throat area)

M = local Mach number

> **Note:** $M = f(A/A^*)$. A can never be less than A^*. For every $A/A^* > 1$ there are **two** solutions: one subsonic and one supersonic; which one occurs depends on the inlet and exit pressures.

A4. Mass Flow Rate Through a Nozzle

Source: Lecture 22.pdf

To find: $\dot{m}/A$ in terms of M and the reservoir (stagnation) conditions p_0, T_0.

Step 1: Start from the definition

$$\frac{\dot{m}}{A} = \rho V = \frac{\rho}{\rho_0}\,\rho_0\,\frac{V}{a}\,a = \frac{\rho}{\rho_0}\,\frac{p_0}{RT_0}\,M\sqrt{\gamma R T}$$

where,

$\dot{m}$ = mass flow rate

V = flow velocity

M = V/a

R = specific gas constant

$p_0,\ T_0$ = stagnation pressure and temperature

ρ_0 = $p_0/(RT_0)$ from the ideal gas law

Step 2: Write $\sqrt{T} = \sqrt{T/T_0}\,\sqrt{T_0}$ and use isentropic relations

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{1}{\gamma - 1}}, \qquad \frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2$$

Step 3: Substitute

$$\frac{\dot{m}}{A} = \frac{p_0 M \sqrt{\gamma/(RT_0)}}{\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{1}{\gamma-1}}\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{1}{2}}}$$

$$\boxed{\frac{\dot{m}}{A} = \frac{p_0 M \sqrt{\gamma/(RT_0)}}{\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}}}$$

where,

$\dot{m}/A$ = mass flux (mass flow per unit area)

A5. Maximum Mass Flow Rate – Choked Flow

Source: Lecture 22.pdf

To find: the Mach number at which $\dot{m}/A$ is maximum, for fixed A, p_0, T_0.

Step 1: Differentiate $\dot{m}/A$ with respect to M and set it to zero

$$\frac{d(\dot{m}/A)}{dM} = 0, \qquad f(M) = M\left(1+kM^2\right)^{-n}, \quad k = \frac{\gamma-1}{2}, \;\; n = \frac{\gamma+1}{2(\gamma-1)}$$

where,

k, n = shorthand constants

$f(M)$ = M-dependent part of $\dot{m}/A$

$$f'(M) = (1+kM^2)^{-n-1}\left[(1+kM^2) - 2nkM^2\right]$$

Since $2nk = \frac{\gamma+1}{2}$, the bracket becomes $1+\frac{\gamma-1}{2}M^2 - \frac{\gamma+1}{2}M^2 = 1 - M^2$. Hence

$$f'(M) = 0 \;\Rightarrow\; \boxed{M = 1}$$

The only place $M = 1$ can occur is the minimum area (throat). So the mass flow through a nozzle is maximum when $M = 1$ at the throat: **choked flow**.

Step 2: Put $M = 1$ and $A = A^*$

$$\boxed{\dot{m}_{max} = A^* p_0 \sqrt{\frac{\gamma}{RT_0}}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}}$$

where,

$\dot{m}_{max}$ = maximum (choked) mass flow rate

A^* = throat area

$$\text{For } \gamma = 1.4: \qquad \dot{m}_{max} = \frac{0.6847\, p_0 A^*}{\sqrt{RT_0}}$$

PART B

Prandtl–Meyer Expansion Waves

Source: Unit 3_A.pdf, Unit 3_B.pdf

B1. Differential Relation Inside an Expansion Wave

Source: Unit 3_A.pdf

Setup: A single Mach wave is produced by an infinitesimal flow deflection $d\theta$. It makes Mach angle μ with the upstream flow. The velocity changes from V to $V + dV$ across it; the velocity component *along* the wave is unchanged.

Step 1: Equal tangential components (from similar triangles ADB and BEF, $BD = BE$)

$$V\cos\mu = (V + dV)\cos(\mu + d\theta)$$

where,

V	=	upstream velocity
$V + dV$	=	velocity after the Mach wave
μ	=	Mach angle
$d\theta$	=	infinitesimal flow deflection

Step 2: Rearrange and expand $\cos(\mu + d\theta)$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos(\mu + d\theta)} = \frac{\cos\mu}{\cos\mu\cos d\theta - \sin\mu\sin d\theta}$$

Step 3: Small-angle assumption: $\sin d\theta \approx d\theta$, $\cos d\theta \approx 1$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu - d\theta\sin\mu} = \frac{1}{1 - d\theta\tan\mu}$$

Step 4: Series expansion $\dfrac{1}{1-x} = 1 + x + x^2 + \cdots\,(x < 1)$, ignoring second-order and higher terms

$$1 + \frac{dV}{V} = 1 + d\theta\tan\mu \;\Rightarrow\; d\theta = \frac{dV/V}{\tan\mu}$$

where,

x	=	$d\theta\tan\mu$ (very small)

Step 5: Mach angle: $\sin\mu = 1/M$ (right triangle with hypotenuse M, opposite side 1, adjacent side $\sqrt{M^2-1}$)

$$\tan\mu = \frac{1}{\sqrt{M^2 - 1}}$$

where,

M	=	Mach number ahead of the wave

$$\boxed{d\theta = \sqrt{M^2 - 1}\,\frac{dV}{V}} \qquad \cdots(1)$$

where,

$d\theta$	=	flow turning angle through the Mach wave
dV/V	=	fractional change of velocity

> **Note:** Equation (1) is the differential equation governing the flow inside an expansion fan. Since $ds = 0$ for each Mach wave, the expansion is isentropic: p_0 and T_0 stay constant.

B2. Prandtl–Meyer Function $\nu(M)$

Source: Unit 3_A.pdf

To find: the total turning angle θ when the Mach number rises from M_1 to M_2 in an expansion fan.

Step 1: Integrate eqn (1) from region 1 ($\theta = 0$, M_1) to region 2 (θ, M_2)

$$\int_0^{\theta} d\theta = \theta = \int_{M_1}^{M_2} \sqrt{M^2 - 1}\,\frac{dV}{V} \qquad \cdots (2)$$

where,

θ = total flow deflection angle

$M_1,\ M_2$ = Mach numbers upstream and downstream of the fan

Step 2: Express dV/V in terms of dM. From $V = Ma$ (take logs and differentiate)

$$\ln V = \ln M + \ln a \ \Rightarrow\ \frac{dV}{V} = \frac{dM}{M} + \frac{da}{a} \qquad \cdots (3)$$

where,

a = local speed of sound

Step 3: Isentropic relation for a

$$\left(\frac{a_0}{a}\right)^2 = \frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}M^2 \ \Rightarrow\ a = a_0\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1/2}$$

where,

a_0 = stagnation speed of sound

$$\frac{da}{a} = -\frac{\gamma - 1}{2}\,M\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1} dM \qquad \cdots (4)$$

Step 4: Put (4) in (3)

$$\frac{dV}{V} = \frac{dM}{M}\left[1 - \frac{\frac{\gamma-1}{2}M^2}{1 + \frac{\gamma-1}{2}M^2}\right] = \frac{1}{1 + \frac{\gamma-1}{2}M^2}\,\frac{dM}{M} \qquad \cdots (5)$$

Step 5: Put (5) in (2) and integrate

$$\theta = \int_{M_1}^{M_2} \frac{\sqrt{M^2 - 1}}{1 + \frac{\gamma-1}{2}M^2}\,\frac{dM}{M} = \left[\sqrt{\tfrac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\tfrac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}\right]_{M_1}^{M_2}$$

$$\boxed{\theta = \nu(M_2) - \nu(M_1)}$$

where,

$\nu(M)$ = Prandtl–Meyer function

$$\boxed{\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\ \tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}\,(M^2-1)}\ -\ \tan^{-1}\sqrt{M^2-1}}$$

where,

γ = ratio of specific heats

$\tan^{-1}$ = inverse tangent, result in radians/degrees

Step 6: Procedure to get the downstream state

1. For given M_1 read $\nu(M_1)$ from the Prandtl–Meyer table.
2. Compute $\nu(M_2) = \nu(M_1) + \theta$.
3. Read M_2 from the table for this $\nu(M_2)$.
4. Expansion is isentropic: $T_{02} = T_{01}$ and $p_{02} = p_{01}$.
5. From isentropic tables read $p_1/p_{01},\ T_1/T_{01}$ at M_1 and $p_2/p_{02},\ T_2/T_{02}$ at M_2.

$$\frac{p_2}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}}, \qquad \frac{T_2}{T_1} = \frac{T_2/T_{02}}{T_1/T_{01}}$$

where,

$p_0,\ T_0$ = stagnation pressure/temperature (same before and after the fan)

Step 7: Mach lines bounding the fan

$$\mu_1 = \sin^{-1}\frac{1}{M_1}\ \ (\text{forward Mach line}), \qquad \mu_2 = \sin^{-1}\frac{1}{M_2}\ \ (\text{rearward Mach line, measured from the downstream flow})$$

where,

$\mu_2 - \theta$ = angle of the rearward Mach line w.r.t. the upstream flow direction

B3. Maximum Turning Angle of a Prandtl–Meyer Expansion

Source: Unit 3_B.pdf

To find: the largest θ obtainable by expansion. From $\theta = \nu(M_2) - \nu(M_1)$, θ is largest when M_1 is smallest and M_2 is largest.

Step 1: Limits: $M_1 = 1$ (sonic start) and $M_2 = \infty$

$$\nu(M_1 = 1) = 0$$

where,

$\nu(1)$ = $\tan^{-1} 0 - \tan^{-1} 0 = 0$

$$\nu(M_2 \to \infty) = \sqrt{\tfrac{\gamma+1}{\gamma-1}} \cdot 90^\circ - 90^\circ$$

where,

$\tan^{-1} \infty$ = 90°

Step 2: Result

$$\boxed{\theta_{max} = \left(\sqrt{\tfrac{\gamma+1}{\gamma-1}} - 1\right) 90^\circ}$$

where,

θ_{max} = maximum possible turning angle

$$\gamma = 1.4: \quad \theta_{max} = (\sqrt{6} - 1)\, 90^\circ = 130.45^\circ$$

PART C

Shock–Expansion Theory: Supersonic Airfoils

Source: Unit 3_C.pdf, Unit 3_D.pdf, Unit 3_E.pdf

C1. Lift and Drag of a Flat Plate at Angle of Attack

Source: Unit 3_C.pdf

Setup: a thin flat plate of chord c at small angle of attack α in supersonic flow $M_1 > 1$ (p_1). The upper surface sees an expansion fan (region 2: $M_2 > M_1$, $p_2 < p_1$). The lower surface sees an oblique shock (region 3: $M_3 < M_1$, $p_3 > p_1$). Forces are per unit span.

Step 1: Resultant pressure force (normal to the plate)

$$R = p_3 c - p_2 c = (p_3 - p_2)\, c$$

where,

R = resultant force normal to plate

p_2 = pressure on upper surface

p_3 = pressure on lower surface

c = plate chord

Step 2: Resolve perpendicular and parallel to the free stream

$$\boxed{L = R\cos\alpha = (p_3 - p_2)\, c\cos\alpha}$$

where,

L = lift (perpendicular to free stream)

α = angle of attack

$$\boxed{D = R\sin\alpha = (p_3 - p_2)\, c\sin\alpha}$$

where,

D = drag (parallel to free stream)

C2. Dynamic Pressure in Terms of Mach Number

Source: Unit 3_D.pdf

Step 1: Start from the definition and multiply and divide by γp_∞

$$q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = \frac{1}{2}\,\frac{\gamma p_\infty}{\gamma p_\infty}\,\rho_\infty V_\infty^2 = \frac{1}{2}\,\gamma p_\infty\,\frac{\rho_\infty}{\gamma p_\infty} V_\infty^2$$

where,

q_∞ = free-stream dynamic pressure

$\rho_\infty,\ p_\infty,\ V_\infty$ = free-stream density, pressure, velocity

Step 2: Use $a_\infty = \sqrt{\gamma p_\infty/\rho_\infty}$

$$q_\infty = \frac{1}{2}\gamma p_\infty \frac{V_\infty^2}{a_\infty^2} \;\Rightarrow\; \boxed{q_\infty = \frac{\gamma}{2}\, p_\infty M_\infty^2}$$

where,

a_∞ = free-stream speed of sound

M_∞ = free-stream Mach number

C3. Lift and Drag Coefficients of a Flat Plate

Source: Unit 3_D.pdf

Step 1: Definitions, with $q_1 = \frac{\gamma}{2} p_1 M_1^2$ and reference area $= c$ (per unit span)

$$C_l = \frac{L}{q_1 c}, \qquad C_d = \frac{D}{q_1 c}$$

where,

$C_l,\ C_d$ = lift and drag coefficients

q_1 = dynamic pressure of the free stream

Step 2: Substitute L, D from C1

$$C_l = \frac{(p_3 - p_2)\, c \cos\alpha}{\frac{\gamma}{2} p_1 M_1^2\, c}$$

$$\boxed{C_l = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha}$$

where,

p_2/p_1 = from expansion fan (1→2), via $\nu(M)$ tables

p_3/p_1 = from oblique shock (1→3), via θ–β–M chart and normal shock tables

$$\boxed{C_d = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\sin\alpha}$$

Step 3: Pressure ratios

$$\frac{p_2}{p_1} = \frac{p_2}{p_{02}} \cdot \frac{p_{02}}{p_{01}} \cdot \frac{p_{01}}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}} \quad (p_{01} = p_{02}, \text{isentropic expansion})$$

$\frac{p_3}{p_1}$ from oblique shock: $M_{n1} = M_1 \sin\beta$

where,

β = oblique shock wave angle

M_{n1} = Mach number component normal to the shock

C4. Diamond Airfoil at Zero Angle of Attack ($\alpha = 0$)

Source: Unit 3_C.pdf

Setup: diamond airfoil with chord *c*, thickness t, side length l and wedge half-angle ε. At $\alpha = 0$ the flow is symmetric. Front faces: oblique shock, pressure $p_2 > p_1$. Rear faces: expansion fan, pressure $p_3 < p_2$. (By symmetry $p_4 = p_2$, $p_5 = p_3$.) Geometry: $l \sin\varepsilon = t/2$, $l \cos\varepsilon = c/2$.

Step 1: Force on each face = pressure × side length ($p\,l$), normal to the face. Sum the components

$$L = -p_2 l\cos\varepsilon + p_2 l\cos\varepsilon - p_3 l\cos\varepsilon + p_3 l\cos\varepsilon \;\Rightarrow\; \boxed{L = 0}$$

where,

L = lift per unit span (perpendicular to free stream)

$p_2 l$ = force on a front face (upper or lower)

$p_3 l$ = force on a rear face (upper or lower)

ε = wedge half-angle

$$D = p_2 l\sin\varepsilon - p_3 l\sin\varepsilon + p_2 l\sin\varepsilon - p_3 l\sin\varepsilon = 2(p_2 - p_3)\, l\sin\varepsilon$$

where,

D = wave drag per unit span (parallel to free stream)

Step 2: Use $l\sin\varepsilon = t/2$

$$\boxed{D = (p_2 - p_3)\, t}$$

where,

t = maximum thickness of the airfoil

> **Note:** In inviscid *incompressible* flow the drag is zero (d'Alembert's paradox). In inviscid supersonic flow the drag is finite: **wave drag**. It arises purely from shock waves (entropy rise and loss of total pressure).

C5. Diamond Airfoil at Angle of Attack $\alpha > \varepsilon$: Lift and Drag Coefficients

Source: Unit 3_E.pdf

Wave pattern (all four faces see a different pressure):

Face	Wave	Turning angle θ
1 → 2 (upper front)	expansion fan	$\theta_2 = \alpha - \varepsilon$
2 → 3 (upper rear)	expansion fan	$\theta_3 = 2\varepsilon$
1 → 4 (lower front)	oblique shock	$\theta_4 = \alpha + \varepsilon$
4 → 5 (lower rear)	expansion fan	$\theta_5 = 2\varepsilon$

Step 1: Normal force N (perpendicular to the chord line); $l\cos\varepsilon = c/2$

$$N = -p_2 l\cos\varepsilon - p_3 l\cos\varepsilon + p_4 l\cos\varepsilon + p_5 l\cos\varepsilon$$

where,

N = force component normal to chord

p_2, p_3 = pressures on upper front/rear faces

p_4, p_5 = pressures on lower front/rear faces

$$\boxed{N = (p_4 + p_5 - p_2 - p_3)\,\frac{c}{2}}$$

where,

c = chord length

Step 2: Axial force A (along the chord line); $l\sin\varepsilon = t/2$

$$A = p_2 l\sin\varepsilon + p_4 l\sin\varepsilon - p_3 l\sin\varepsilon - p_5 l\sin\varepsilon$$

where,

A = force component along chord

$$\boxed{A = (p_2 + p_4 - p_3 - p_5)\,\frac{t}{2}}$$

where,

t = maximum thickness

Step 3: Rotate to the wind axes (free-stream direction)

$$\boxed{L = N\cos\alpha - A\sin\alpha}, \qquad \boxed{D = N\sin\alpha + A\cos\alpha}$$

where,

L, D = lift and drag per unit span

α = angle of attack

Step 4: Coefficients, with $q_1 = \frac{\gamma}{2}p_1 M_1^2$ and reference area c

$$C_l = \frac{L}{q_1 c} = \frac{2}{\gamma M_1^2 c}\left[\left(\tfrac{p_4}{p_1} + \tfrac{p_5}{p_1} - \tfrac{p_2}{p_1} - \tfrac{p_3}{p_1}\right)\frac{c}{2}\cos\alpha - \left(\tfrac{p_2}{p_1} + \tfrac{p_4}{p_1} - \tfrac{p_3}{p_1} - \tfrac{p_5}{p_1}\right)\frac{t}{2}\sin\alpha\right]$$

$$\boxed{C_l = \frac{1}{\gamma M_1^2}\left[\left(\tfrac{p_4}{p_1} + \tfrac{p_5}{p_1} - \tfrac{p_2}{p_1} - \tfrac{p_3}{p_1}\right)\cos\alpha - \left(\tfrac{p_2}{p_1} + \tfrac{p_4}{p_1} - \tfrac{p_3}{p_1} - \tfrac{p_5}{p_1}\right)\frac{t}{c}\sin\alpha\right]}$$

where,

t/c = thickness ratio, $= \tan\varepsilon$

p_i/p_1 = pressure ratio of face i to free stream

$$\boxed{C_d = \frac{1}{\gamma M_1^2}\left[\left(\tfrac{p_4}{p_1} + \tfrac{p_5}{p_1} - \tfrac{p_2}{p_1} - \tfrac{p_3}{p_1}\right)\sin\alpha + \left(\tfrac{p_2}{p_1} + \tfrac{p_4}{p_1} - \tfrac{p_3}{p_1} - \tfrac{p_5}{p_1}\right)\frac{t}{c}\cos\alpha\right]}$$

Step 5: Each pressure ratio is found with the shock/expansion tools

$$\frac{p_2}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}}, \quad \frac{p_3}{p_1} = \frac{p_3/p_{03}}{p_1/p_{01}}\ \ (p_{01} = p_{02} = p_{03}), \quad \frac{p_5}{p_1} = \frac{p_5}{p_{05}}\cdot\frac{p_{04}}{p_{01}}\cdot\frac{p_{01}}{p_1}\ \ (p_{04} = p_{05})$$

where,

p_{0i} = stagnation pressure in region i

p_{04}/p_{01} = stagnation-pressure ratio across the oblique shock (from normal shock tables with M_{n1})

$$\nu(M_2) = \nu(M_1) + \theta_2, \quad \nu(M_3) = \nu(M_2) + \theta_3, \quad M_4 = \frac{M_{n4}}{\sin(\beta_4 - \theta_4)}, \quad \nu(M_5) = \nu(M_4) + \theta_5$$

where,

β_4 = shock angle for the lower front face

M_{n4} = normal Mach number downstream of the shock

PART D

One-Dimensional Flow with Heat Addition (Rayleigh Flow)

Source: Unit 4_Rayleigh Flow.pdf

D1. Governing Equations and Effect of Heat Addition

Source: Unit 4_Rayleigh Flow.pdf

Setup: steady 1-D flow in a constant-area duct ($A_1 = A_2 = A$), no friction, with heat q (per unit mass) added between sections 1 and 2. Gas is calorically perfect.

Step 1: Basic equations

$$\rho_1 u_1 = \rho_2 u_2$$

where,

ρ = density

u = velocity

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

where,

p = static pressure

$$q + h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

where,

q = heat added per unit mass (J/kg)

h = specific enthalpy

Step 2: Energy equation with $h = c_p T$

$$q = \left(c_p T_2 + \frac{u_2^2}{2}\right) - \left(c_p T_1 + \frac{u_1^2}{2}\right) = c_p\left(T_2 + \frac{u_2^2}{2c_p}\right) - c_p\left(T_1 + \frac{u_1^2}{2c_p}\right)$$

where,

c_p = specific heat at constant pressure

T = static temperature

Since $T_0 = T + \frac{u^2}{2c_p}$ (total temperature):

$$\boxed{q = c_p\,(T_{02} - T_{01})}$$

where,

$T_{01},\ T_{02}$ = stagnation temperatures at sections 1 and 2

> **Note:** For adiabatic flow $T_{01} = T_{02}$. Heat addition directly changes the total temperature: heating raises T_0, cooling lowers it. We now find the property ratios in terms of M_1 and M_2.

D2. Pressure Ratio p_2/p_1 in Rayleigh Flow

Source: Unit 4_Rayleigh Flow.pdf

Step 1: Momentum equation

$$p_2 - p_1 = \rho_1 u_1^2 - \rho_2 u_2^2$$

Step 2: Write ρu^2 in terms of M: since $u = Ma$ and $a = \sqrt{\gamma p/\rho}$

$$\rho u^2 = \rho a^2 M^2 = \rho\,\frac{\gamma p}{\rho}\,M^2 = \gamma p M^2$$

where,

a = speed of sound

M = Mach number

γ = ratio of specific heats

Step 3: Substitute

$$p_2 - p_1 = \gamma p_1 M_1^2 - \gamma p_2 M_2^2 \;\Rightarrow\; p_2(1 + \gamma M_2^2) = p_1(1 + \gamma M_1^2)$$

$$\boxed{\frac{p_2}{p_1} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2}} \qquad \cdots (1)$$

D3. Temperature Ratio T_2/T_1

Source: Unit 4_Rayleigh Flow.pdf

Step 1: Ideal gas law $T = p/(\rho R)$ and continuity $\rho_1 u_1 = \rho_2 u_2$

$$\frac{T_2}{T_1} = \frac{p_2}{p_1}\frac{\rho_1}{\rho_2} = \frac{p_2}{p_1}\frac{u_2}{u_1}$$

where,

R = specific gas constant

Step 2: Velocity ratio, using $u = Ma = M\sqrt{\gamma RT}$

$$\frac{u_2}{u_1} = \frac{M_2 a_2}{M_1 a_1} = \frac{M_2}{M_1}\left(\frac{T_2}{T_1}\right)^{1/2}$$

Step 3: Combine and cancel one power of $\sqrt{T_2/T_1}$

$$\left(\frac{T_2}{T_1}\right)^{1/2} = \frac{p_2}{p_1}\frac{M_2}{M_1}$$

Using eqn (1):

$$\boxed{\frac{T_2}{T_1} = \left(\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right)^2 \left(\frac{M_2}{M_1}\right)^2} \qquad \cdots (2)$$

D4. Density Ratio ρ_2/ρ_1

Source: Unit 4_Rayleigh Flow.pdf

Step 1: Ideal gas law

$$\frac{\rho_2}{\rho_1} = \frac{p_2}{p_1}\frac{T_1}{T_2}$$

Step 2: Substitute (1) and (2)

$$\boxed{\frac{\rho_2}{\rho_1} = \left(\frac{1+\gamma M_2^2}{1+\gamma M_1^2}\right)\left(\frac{M_1}{M_2}\right)^2} \qquad \cdots (3)$$

D5. Stagnation Pressure Ratio p_{02}/p_{01}

Source: Unit 4_Rayleigh Flow.pdf

Step 1: Break it into known ratios

$$\frac{p_{02}}{p_{01}} = \frac{p_{02}}{p_2} \times \frac{p_2}{p_1} \times \frac{p_1}{p_{01}}$$

where,

p_0 = stagnation (total) pressure

Step 2: Isentropic relation $p_0/p = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\gamma/(\gamma-1)}$ at each section, and eqn (1)

$$\boxed{\frac{p_{02}}{p_{01}} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2}\left[\frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}\right]^{\frac{\gamma}{\gamma-1}}} \qquad \cdots (4)$$

D6. Stagnation Temperature Ratio T_{02}/T_{01}

Source: Unit 4_Rayleigh Flow.pdf

Step 1: Break it into known ratios

$$\frac{T_{02}}{T_{01}} = \frac{T_{02}}{T_2} \times \frac{T_2}{T_1} \times \frac{T_1}{T_{01}}$$

where,

T_0 = stagnation (total) temperature

Step 2: Isentropic relation $T_0/T = 1 + \frac{\gamma-1}{2}M^2$ at each section, and eqn (2)

$$\boxed{\frac{T_{02}}{T_{01}} = \left[\frac{1+\gamma M_1^2}{1+\gamma M_2^2}\right]^2 \left(\frac{M_2}{M_1}\right)^2 \frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}} \qquad \cdots(5)$$

> **Note:** Equations (1)–(5): $p_2/p_1,\ T_2/T_1,\ \rho_2/\rho_1,\ p_{02}/p_{01},\ T_{02}/T_{01}$ are all functions of (M_1, M_2) only. Given q, M_2 would have to be found from (5) by trial and error (no direct solution), so the sonic reference method below is used instead.

D7. Sonic Reference Conditions: $q = q_1^* - q_2^*$ and $p_1^* = p_2^*$

Source: Unit 4_Rayleigh Flow.pdf

Idea: for a given flow, imagine heat is added until $M = 1$. Properties there are $p^*, T^*, \rho^*, p_0^*, T_0^*$ (star = sonic reference).

Step 1: Heat needed to reach $M = 1$ from state 1, and from state 2

$$q_1^* = c_p(T_0^* - T_{01}), \qquad q_2^* = c_p(T_0^* - T_{02})$$

where,

q_1^* = heat per unit mass needed to choke the flow from state 1

q_2^* = heat per unit mass needed to choke the flow from state 2

Step 2: Subtract

$$\boxed{q = c_p(T_{02} - T_{01}) = q_1^* - q_2^*}$$

where,

q = heat actually added between 1 and 2

Step 3: Show the sonic state is the same for both: apply eqn (1) between each state and the sonic state

$$\text{(I)}\ \ 1 \to *: \quad \frac{p_1^*}{p_1} = \frac{1+\gamma M_1^2}{1+\gamma} \qquad \text{(II)}\ \ 2 \to *: \quad \frac{p_2^*}{p_2} = \frac{1+\gamma M_2^2}{1+\gamma}$$

$$\frac{\text{(I)}}{\text{(II)}}: \quad \frac{p_2}{p_1}\cdot\frac{p_1^*}{p_2^*} = \frac{1+\gamma M_1^2}{1+\gamma M_2^2}$$

Comparing with eqn (1), $p_2/p_1 = (1+\gamma M_1^2)/(1+\gamma M_2^2)$:

$$\boxed{p_1^* = p_2^*}$$

where,

p^* = sonic reference pressure

> **Note:** Sonic reference conditions $p^*, T^*, \rho^*, p_0^*, T_0^*$ are **constants** for the whole Rayleigh flow, however the local properties change.

D8. Rayleigh Flow Property Ratios Relative to Sonic Conditions

Source: Unit 4_Rayleigh Flow.pdf

Method: put state 1 = local state (p, M) and state 2 = sonic state (p^*, $M = 1$) in eqns (1)–(5).

Step 1: Static pressure, from (1)

$$\frac{p^*}{p} = \frac{1+\gamma M^2}{1+\gamma} \ \Rightarrow\ \boxed{\frac{p}{p^*} = \frac{1+\gamma}{1+\gamma M^2}}$$

where,

p/p^* = static pressure relative to sonic value

M = local Mach number

Step 2: Static temperature, from (2)

$$\frac{T^*}{T} = \left(\frac{1+\gamma M^2}{1+\gamma}\right)^2 \frac{1}{M^2} \ \Rightarrow\ \boxed{\frac{T}{T^*} = M^2\left(\frac{1+\gamma}{1+\gamma M^2}\right)^2}$$

where,

T/T^* = static temperature relative to sonic value

Step 3: Density, from (3) (or $\rho/\rho^* = (p/p^*)(T^*/T)$)

$$\boxed{\frac{\rho}{\rho^*} = \frac{1}{M^2}\left(\frac{1+\gamma M^2}{1+\gamma}\right)}$$

where,

ρ/ρ^* = density relative to sonic value

Step 4: Stagnation pressure: $\dfrac{p_0}{p_0^*} = \dfrac{p_0}{p} \cdot \dfrac{p}{p^*} \cdot \dfrac{p^*}{p_0^*}$

$$\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}, \qquad \frac{p^*}{p_0^*} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\boxed{\frac{p_0}{p_0^*} = \frac{1+\gamma}{1+\gamma M^2}\left[\frac{2+(\gamma-1)M^2}{\gamma+1}\right]^{\frac{\gamma}{\gamma-1}}}$$

where,

p_0/p_0^* = stagnation pressure relative to sonic value

Step 5: Stagnation temperature: $\dfrac{T_0}{T_0^*} = \dfrac{T_0}{T} \cdot \dfrac{T}{T^*} \cdot \dfrac{T^*}{T_0^*}$

$$\frac{T_0}{T} = 1 + \frac{\gamma-1}{2}M^2, \qquad \frac{T^*}{T_0^*} = \frac{2}{\gamma+1}$$

$$\boxed{\frac{T_0}{T_0^*} = \frac{(\gamma+1)M^2}{(1+\gamma M^2)^2}\left[2+(\gamma-1)M^2\right]}$$

where,

T_0/T_0^* = stagnation temperature relative to sonic value

> **Note:** These five ratios are tabulated as Rayleigh-flow tables (Table 8.2, $\gamma = 1.4$). Use: $T_{02} = T_{01} + q/c_p$, then T_{02}/T_0^* (with T_0^* from state 1) gives M_2 directly, and $p_2/p_1 = (p_2/p^*)(p^*/p_1)$ etc.

D9. Choking in Rayleigh Flow and Effect of Heat Addition

Source: Unit 4_Rayleigh Flow.pdf

Step 1: Maximum heat that can be added without altering the inlet condition

$$\boxed{q_1^* = c_p\,(T_0^* - T_{01})}$$

where,

q_1^* = heat that brings the exit Mach number to 1 (choked)

T_0^* = sonic stagnation temperature, from T_{01}/T_0^* at M_1

Step 2: Effect of heat addition ($q > 0$) on flow properties

Property	$M_1 < 1$ **(subsonic)**	$M_1 > 1$ **(supersonic)**
Mach number	increases, $M_2 > M_1$	decreases, $M_2 < M_1$
Static pressure	decreases, $p_2 < p_1$	increases, $p_2 > p_1$
Static temperature	increases, $T_2 > T_1$	increases, $T_2 > T_1$
Stagnation pressure	decreases, $p_{02} < p_{01}$	decreases, $p_{02} < p_{01}$
Stagnation temperature	increases, $T_{02} > T_{01}$	increases, $T_{02} > T_{01}$

Step 3: What happens when $q > q_1^*$

Inlet	**Result for** $q > q_1^*$
$M_1 < 1$	Exit stays choked ($M_2 = 1$); the inlet condition adjusts: M_1 decreases and mass flow reduces.
$M_1 > 1$	A normal shock forms upstream of the inlet and makes the entry flow subsonic; the exit stays choked.

> **Note:** On the Rayleigh curve (Mollier h–s diagram) the sonic point a is the point of maximum entropy. The upper branch is subsonic and the lower branch is supersonic; heating moves the state toward a on both branches.